

PAC-Former — mandatory PAC-interaction tokenizer + tri-axial encoder

TUEV supervised, seed 0: $\kappa = 0.5493$ vs. raw-token backbone 0.4679 (AGENT.md §13.43-I); both 1,635,734 parameters

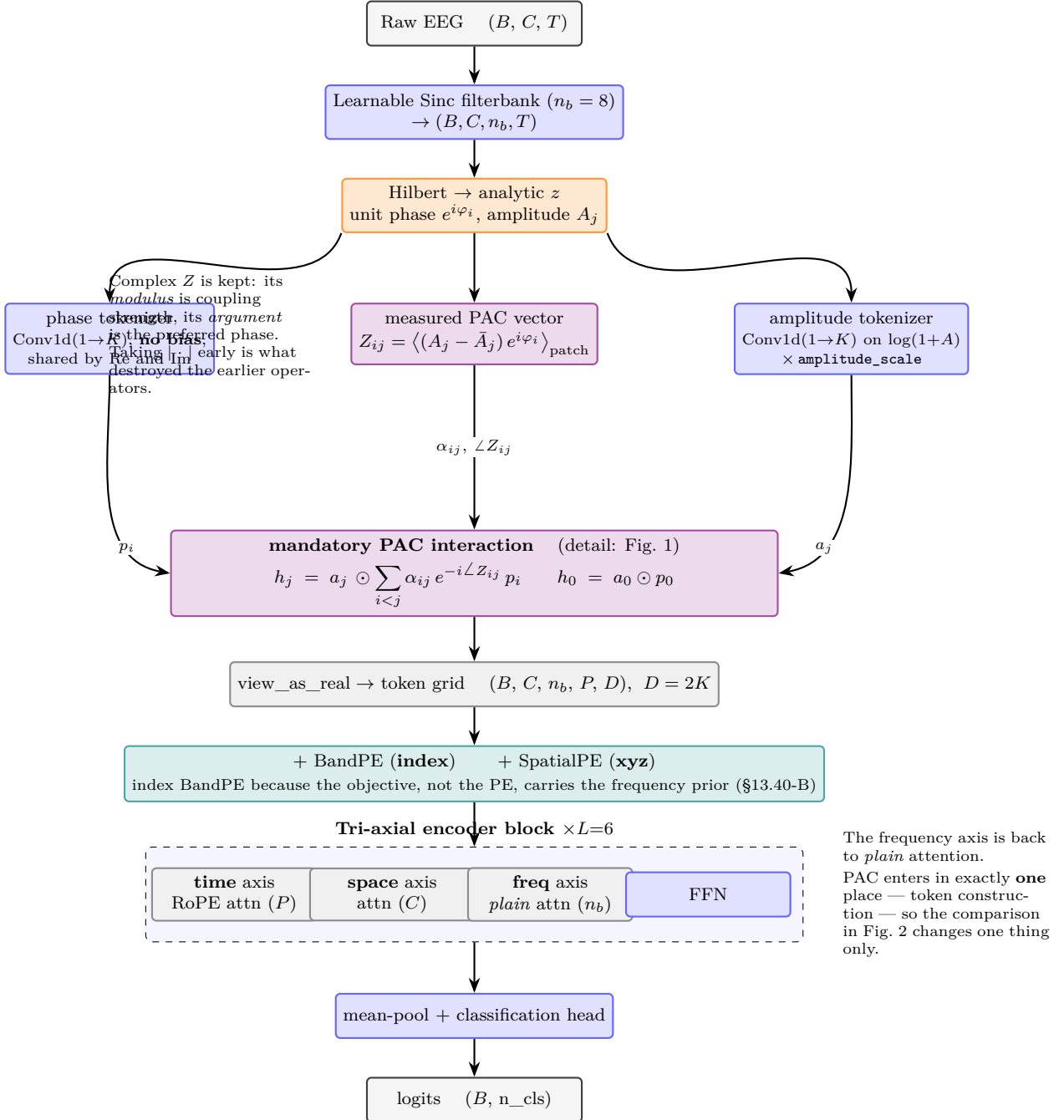
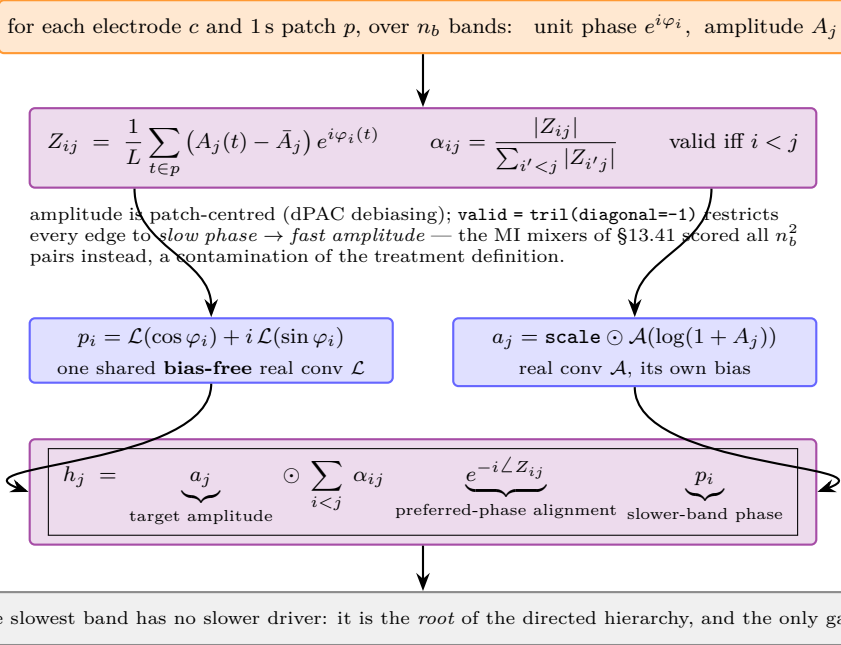


Fig. 1 — the mandatory, phase-reference-invariant PAC token



Why the prior cannot be routed around. There is no raw high-band token beside h_j , no `pac_scale`, no gate, no PAC branch, no auxiliary loss. Every sample-dependent feature reaching the encoder for $j > 0$ contains *both* the target amplitude and an aligned slower-band phase. BandPE/SpatialPE are constants and cannot carry sample-specific EEG around the product. This is the §13.18 lesson applied structurally: a prior with a free path beside it gets optimised away, so we removed the path.

Phase-reference (gauge) invariance.

Shift band i 's phase origin by an arbitrary δ_i :

$$\begin{aligned} \mathcal{L} \text{ is real-linear and bias-free} &\Rightarrow p_i \mapsto e^{i\delta_i} p_i, \\ Z_{ij} \mapsto e^{i\delta_i} Z_{ij} &\Rightarrow e^{-i\angle Z_{ij}} \mapsto e^{-i\delta_i} e^{-i\angle Z_{ij}} \end{aligned}$$

the two rotations cancel exactly, so $h_{j>0}$ is **unchanged**. Verified numerically: $\max |\Delta| = 8.3 \times 10^{-7}$ over bands 1..7 (`scratchpad/verify_pacint_configs.py`).

This is a property of the operator, not something training must discover — and it is the concrete difference from the old `FreqPhaseSteered` mixer, which rotated learned real feature pairs that had no defined phase reference.

Fig. 2 — the three parameter-matched arms (TUEV, seed 0, 1,635,734 params each)

measured	$\alpha_{ij} = \frac{ Z_{ij} }{\sum_j Z_{.j} }$, phase factor $e^{-i\angle Z_{ij}}$	$\kappa = 0.5493$	{ +0.0394 preferred-phase owner- ship +0.0365 measured PAC magni- tude +0.0055 — tie the product topology alone is worth nothing
scramble	same $ Z_{ij} $, but <i>which edge owns which preferred phase</i> is freshly permuted every forward	$\kappa = 0.5099$	
uniform	$\alpha_{ij} = 1/ \{i < j\} $, no phase factor — the product topology survives, the measurement does not	$\kappa = 0.4734$	
raw-token baseline	per-(c, b) Conv1d on the filtered waveform; no interaction at all	$\kappa = 0.4679$	

Reading. The decisive control is **uniform**, not **measured**. It keeps the entire mandatory amplitude $\times$ slower-phase structure and only swaps the measured coefficients for a uniform average — and it *ties* the raw baseline. So the gain is not the multiplicative token structure; it is the measured quantity poured into it. The two contrasts above then split that gain into two nearly equal, separately attributable halves.

Not yet established. **measured**–**scramble** = +0.0394 is the only evidence that *preferred phase specifically* matters; it is barely $2\times$ the 0.02 single-seed noise line, and **scramble** redraws its permutation every forward while **measured** does not (val std 0.0466 vs 0.0382), so part of it may be injected stochasticity. Needs seeds 1, 2 and a *deterministic* phase-destruction arm (real α_{ij} , no $e^{-i\angle Z_{ij}}$) before the preferred-phase claim is written.